Erfan Nasirzadeh Orang

Machine Learning · Uncertainty Quantification · Human-Centered Computing

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Education

Master of Computing and Information Systems (MCIS) Aug 2026 – Present
Youngstown State University, OH
Advisor: Dr. Zack While · Thesis Based
Fully Funded Graduate Research Assistantship

Computer Science, Bachelor of Science with Honors Aug 2022 – May 2026
Youngstown State University, OH
Minors: Mathematics, Data Analytics · Honors Scholar with Distinction
GPA: 3.50/4.00 cumulative, 3.73/4.00 major
Dean's List (4 semesters) · President's List (2 semesters)

Research Experience

Graduate Research Assistant, Computer Science Dept., YSU May 2026–Present
Advisors: Dr. Zack While
Projects: HCI/fMRI Neural Encoding

- Evaluated Meta's Tribe V2 neural encoding model to replicate fMRI visualization studies, successfully mapping model-predicted cortical responses against human behavioral baselines in visual-processing regions (Accepted, IEEE VIS '26).

Projects: Physics-Informed ML Surrogates

- Trained predictive ML surrogate models (XGBoost, Random Forest, GPR) on URANS CFD datasets to evaluate turbulence sensitivity on H-type Darrieus vertical-axis wind turbines (Under Review, *Part A: Journal of Power and Energy*).
- Implemented SHapley Additive exPlanations (SHAP) and nested cross-validation to interpret turbine predictions and extract spatially resolved uncertainty estimates.
- Developed machine learning surrogate models to optimize Al_2O_3–TiO_2/Water hybrid nanofluid battery thermal management systems (Accepted, *JTAC*).

Undergraduate Data Science Researcher, Jan–May 2026
Advisors: Dr. Jeffrey Coldren *YSU Data Mine & Purdue University*
Project: Connections Build Student Success and Retention, YSU Student Experience

- Analyzed relationships between academic variables and retention for 1,935 students using Ohio Super
- Trained and tested multiple ML models to predict student success, engineering new features to capture engagement patterns and presenting findings to university leadership.

REU Research Intern, Williams College, Williamstown, MA Jun–Aug 2024
Project: Generative Machine Learning Models for Data Assimilation

- Collaborated with mentors from Caltech, CUNY, and NYU to develop a conditional sampling algorithm for improved weather and financial forecasting.
- Applied optimal transport theory and machine learning to analyze atmospheric datasets, achieving a 12% improvement in predictive accuracy for weather models.

Imaging Lab Technician Intern, Depts. of Physics & Biology, YSU Aug–Dec 2023
Project: Spectral Fingerprint of Laser Emission from Rh6G-Infused Peafowl Tail Feathers

- Trained and certified to operate Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), and Transmission Electron Microscopy (TEM), ensuring proper sample preparation, imaging procedures, and routine maintenance.

- Assisted a Master's student whose thesis focused on a novel neurological disease treatment using gold nanoparticles, performed nanoscale imaging (AFM, SEM, TEM) to characterize nanoparticle morphology, size distribution, and dispersion for the project.
- Collaborated with researchers from Florida Polytechnic University on optical measurements of rhodamine 6G–infused peafowl feather samples; this work led to co-authorship.

Lab Research Assistant, Physics & Astronomy Department, YSU Oct 2022–May 2025
Project: Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency

- Conducted laser-based quantum optics experiments and performed theoretical modeling/signal processing using Python and MATLAB, contributing to a co-authored publication.

Technical Skills

Languages: Python, MATLAB, R, C, SQL, PHP
ML / Data: scikit-learn, XGBoost, Random Forest, Gaussian Process Regression, SHAP, nested cross-validation, feature engineering, pandas
Tools: Git/GitHub, LaTeX, MySQL

Publications

Peer-Reviewed Journal Articles

J. Gorkos, K. Grenzig, **E. N. Orang**, V. Thomas, D. Tighe, and M. Crescimanno. *Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency. Journal of the Optical Society of America B*, Vol. 42, No. 2, pp. 404–411, Feb 2025. [DOI: 10.1364/JOSAB.544782]

A. Fiorito, D. R. Sheffield, H. Liu, **E. N. Orang**, and N. J. Dawson. *Spectral fingerprint of laser emission from rhodamine 6g infused male Indian Peafowl tail feathers. Scientific Reports*, Vol. 15, Article 20938, 2025. [DOI: 10.1038/s41598-025-04039-8]

A. Ulus, I. Uluocak, **E. N. Orang**, N. Vakili, and K. Choo. *Machine learning surrogates for Al_2O_3–TiO_2/water hybrid nanofluid battery thermal management under variable heat generation. Journal of Thermal Analysis and Calorimetry*, 2026. [DOI: 10.1007/s10973-026-16179-8]

Under Review

A. Ulus, I. Uluocak, **E. N. Orang**, N. Vakili, and K. Choo. *Operating-regime dependence of turbulence effects on an H-type Darrieus vertical-axis wind turbine using uncertainty-aware surrogate modeling. Part A: Journal of Power and Energy*, (Under Review, Submitted Aug 2026).

Conference & Workshop Papers

E. N. Orang and Z. While. *Can a Neural Encoding Model Replicate an fMRI Visualization Study? VISxVISION Workshop, IEEE VIS '26*, Boston, MA, Nov 2026. [`https://ieeevis.org/year/2026/program/paper/3b9c6999-3420-49c7-a24f-92a1edc21f2a/`] [arXiv:2609.15685] [Supplemental: `https://osf.io/8a96x/`]

Oral Presentations

Spectral fingerprint of laser emission from rhodamine 6g infused male Indian Peafowl tail feathers. Oral presentation by **E. N. Orang**, Fall 2025 Joint Meeting of the Eastern Great Lakes Section and MIAAPT, Ypsilanti, Michigan, October 2025.
[*Link to Session Abstract*]

A Mathematical Model of Foot Traffic-Induced Wear on Historical Staircases: A Solu-

tion to the COMAP Mathematical Contest in Modeling Problem. Oral presentation by **E. N. Orang**. Pi Mu Epsilon Honor Society Annual Conference, Youngstown State University, March 2025.
[*Link to Session Abstract*]

Poster Presentations

Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency. Poster presentation by **E. N. Orang** at the 59th Annual Conference of the National Collegiate Honors Council (NCHC), San Diego, California, September 2025.
[*Link to Session Abstract*]

Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency. Poster presentation by **E. N. Orang** at the 55th Annual Meeting of the APS Division of Atomic, Molecular, and Optical Physics (DAMOP), Session S00.38, Fort Worth, Texas, June 2024.
[*Link to Session Abstract*]

Press Coverage

"Peacock feathers can be lasers", Science Magazine (AAAS), July 2025.[*Link to Article*]

"Study Detects Lasers in Peacock Tail Feathers", The Business Journal, August 2025.
[*Link to Article*]

"Florida Poly student research uncovers laser light from peacock feathers", Florida Polytechnic University News, August 2025.[*Link to Article*]

Certifications & Projects

COMAP Mathematical Contest in Modeling (MCM) 2025 Successful Participant
Problem A: The Testing Time Of The Constant Wear On Historical Stairs
Worked in a three-student team to develop a mathematical model for Problem A; recognized as a Successful Participant in the 2025 COMAP competition.
[Link to YSU COMAP Modeling Contest]

Academic Leadership

Peer Tutor – CS & Mathematics, RASC, YSU Aug 2025–May 2026

- Tutor lower- and upper-division computer science courses, including Fundamental Programming and Problem Solving I–II, Data Structures and Objects, Development of Databases (SQL), E-Commerce Programming, Advanced Object-Oriented Programming, and Introduction to Discrete Structures.
- Provide additional support in Linear Algebra and Matrix Theory, Probability and Statistics, emphasizing problem-solving strategies and exam preparation.

Honors Student Teaching Assistant, Sokolov Honors College, YSU Aug 2024–May 2026

- Provided academic guidance to 65+ honors students each academic year, assisting with curriculum planning, Honors Track requirements, and university resources.
- Led weekly review sessions and facilitated group discussions to reinforce core concepts and encourage collaborative learning.

Community Engagement

International Orientation Leader Aug 2023–May 2026
International Programs Office, YSU

- Co-organized and led campus tours and info-sessions for 1,200+ incoming students.

Community Fellow, **Event Organizer**, **Honors College Liaison** Sep 2023–May 2024
The Roger & Gloria Jones Children's Center for Science & Technology, Youngstown, OH

- Co-designed workshops (robotics demos, circuitry labs, coding challenges) with OH WOW! educators and Honors College faculty, aligning community goals with academic objectives.

Iranian Youth Society, Member & Social Relations Officer Sep 2022–Present
Youngstown State University

- Coordinated peer support programs to help newly arrived students adapt quickly to a new environment and maintain continuous personal and academic growth.